

[14X030] Introduction to Computational Finance

Exercise #11

May 12, 2026

General instructions

Each student is expected to upload on Moodle a **zip** file named as *firstname_lastname*, containing the followings:

- A report in **pdf** format to answer exercise questions.
- The **code** used to generate the results of the report.
- If preferred, submit a unique **Jupyter notebook** with both code and report explanations.

Deeper instructions

- **Report:**
 - Answers to the TP questions. Figures and numerical results are necessary but **strictly not** sufficient: provide your analysis/comments as well.
 - Example: if your results surprise you, explain why and explain what the expected result was.
 - **Pay attention** to the presentation: axis labels, legend, etc.
- **Code:**
 - Implementation questions + code used to produce the results and figures in the report.
 - Programming language of your choice.
 - Do not use ChatGPT, or any other LLM/AI-assisted tools while writing down your code. Try by yourself, otherwise it will be counted as failed.
 - I **strongly recommend Python/Julia**: I won't be able to assist you in the same way if you use a different language.

Deadline

Any questions to: [Lorenzo Bini](#).

Upload on Moodle due by : **May 18, 2026 at 11:59 pm**

Bonds

Suppose we have a bond with 3 years maturity, a face value of 100\$ and coupons of 10% paid semi-annually:

time [years]	coupon [\$]
0.5	5
1	5
1.5	5
2	5
2.5	5
3	5

- What is the required yield to sell the bond at par?

Interest curve

Consider we have some bonds with a coupon paid semi-annually (Fig. 1) and some other with a coupon paid annually (Fig. 2).

- Using the bootstrap algorithm seen during the course, compute the zero-coupon rate for each maturity.
- Draw the graph of the zero-coupon rate against the maturity. What do you observe?

Duration and Interest Rate Sensitivity

Lastly, you will compute the Macaulay and modified durations of selected bonds using the zero-coupon rates obtained from bootstrapping. Then, you will analyze how bond prices respond to small parallel shifts in the yield curve.

1. Select Two Bonds: Choose one bond from 1 (semi-annual coupons) and one bond from 2 (annual coupons).

Principal	Maturity (month)	Coupons (%)	Price
100	1	0	99.80
100	2	0	99.60
100	3	0	99.40
100	6	3	100.27
100	12	4	101.57

Figure 1: Bonds with a coupon paid semi-annually

Principal	Maturity (year)	Coupons (%)	Price
100	2	4	103.21
100	3	4	104.85
100	4	4	106.36
100	5	4	107.77
100	7	0	84.48
100	10	0	77.72

Figure 2: Bonds with a coupon paid annually

2. Compute Durations:

- Using the zero-coupon rates computed in Exercise 2, calculate the present value of each cash flow.
- Compute the **Macauley duration** as the weighted average time to receive the bond's cash flows:

$$D_M = \sum_i \left(\frac{PV_i}{P} \cdot t_i \right),$$

where PV_i is the present value of the cash flow at time t_i , and P is the total bond price.

- Then compute the **modified duration**, which adjusts for the interest rate level:

$$D_{\text{mod}} = \frac{D_M}{1 + r},$$

where r is the yield used to discount the cash flows.

3. Simulate a parallel upward shift of 10 basis points (0.001) in the entire zero-coupon curve.
4. Recompute the bond price under the shifted curve.
5. Compare the predicted price change using modified duration with the actual price change from repricing.
6. How well does modified duration approximate the price change? Which bond is more sensitive to rate changes and why?